WGNI FARM

STANDARD OPERATING PROCEDURE

OPEN FIELD AND ORCHARD OPERATIONS

FIELD	DETAIL
Document Title	Open Field and Orchard Operations Standard Operating Procedure
Covers	Seed Beds 1, 2 and 3 (Sweet Corn, Tomatoes and future crops) and Orchard (Mango, Orange and future trees)
Location	WGNI Farm, Abeokuta, Nigeria
Document Authority	CEO, WGNI Farm
Applies To	Field Worker, Agronomist, Farm Manager
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DEFINITIONS

Mulch Film: Black polythene sheeting laid over raised planting beds to suppress weeds, retain soil moisture, regulate soil temperature, and reduce soil-borne disease splash onto the plant. Plants are established through holes cut directly into the mulch film.

Drip Emitter: The individual outlet point on a drip irrigation line through which water is delivered directly to each plant at the base of the stem.

Transplanting: The process of moving a seedling raised in the nursery into its final growing position in the open field bed.

Fertigation: The process of delivering dissolved nutrients to plants through the drip irrigation system simultaneously with water.

Inter-row Space: The exposed soil walkway between two adjacent planting beds. This area is not covered by mulch film and requires regular manual weeding.

Staking: The process of driving a firm support stake into the ground beside a plant and tying the plant to it to prevent collapse under the weight of fruit or from wind.

Sucker: A lateral shoot that grows from the junction between the main stem and a leaf stalk on tomato plants. Suckers divert energy from fruit production and are removed as directed by the Agronomist.

Roguing: The removal and destruction of diseased, infected, or off-type plants from the field to prevent the spread of disease or genetic contamination of the crop stand.

Orchard: A defined area of land where fruit trees are planted, managed, and maintained for commercial fruit production over multiple seasons.

Canopy: The upper layer of foliage formed by the branches and leaves of a tree. Managing the canopy through pruning controls light penetration, air circulation, and fruit development.

Base Circle: The area of soil directly around the base of a tree trunk, maintained as bare or mulched ground to prevent competition from grass and weeds and to protect the root zone.

Graft Union: The swollen join between the rootstock and the scion on a grafted tree seedling. The graft union must always remain above the soil surface after planting.

Formative Pruning: Pruning carried out on young trees during the first three years to shape the tree into a productive and well-structured canopy before commercial fruiting begins.

SECTION 1 - ACCESS AND SECURITY

1.1 Open Field Access

Assigned field workers, the Agronomist, and the Farm Manager are authorised to work in and access the seed beds at all times during working hours. Visitors and buyers may observe from the field perimeter. They must not enter active beds without Farm Manager authorisation. No visitor may handle plants, touch drip lines, or disturb bed structures without explicit Agronomist instruction.

1.2 Orchard Access

- The orchard area is partially demarcated. Until full demarcation is complete, the Agronomist marks the orchard boundary clearly and all workers are briefed on the boundary before any work begins.
- No vehicle may enter the orchard area without Farm Manager authorisation. Vehicle movement near tree root zones causes soil compaction and root damage.
- No livestock may access the orchard area at any time.

1.3 Security

- No harvested produce, plant material, or equipment may leave the field without authorisation from the Agronomist or Farm Manager.
- Any evidence of theft, vandalism, unauthorised entry, or encroachment must be reported to the Farm Manager immediately.
- All tools and equipment must be returned to the designated store at the end of every working day. No tool is left unattended in the open field overnight.

SECTION 2 - BED PREPARATION STANDARDS

2.1 Bed Structure Standard

All open field seed beds at WGNI Farm are raised beds covered with black mulch film. The following standards apply to every bed before any crop is planted:

- Bed width: 90 to 100 cm across the top of the raised bed.
- Bed height: 15 to 20 cm above the surrounding ground level.
- Walkway width between beds: minimum 50 cm to allow comfortable worker access.
- Bed surface must be firm, level, and free of large clods, stones, crop residue, and debris before mulch film is applied.

2.2 Soil Preparation

- Clear the bed area of all vegetation, roots, and previous crop residue before bed formation.
- Till the soil to a depth of at least 25 to 30 cm to loosen compaction and improve root penetration.
- Incorporate organic matter or basal fertiliser as directed by the Agronomist before the bed is shaped.
- Shape the bed by mounding the tilled soil into the correct width and height.
- Smooth and firm the bed surface by hand before drip line installation and mulch film application.

2.3 Drip Line Installation

- Lay drip lines along the length of each bed before the mulch film is applied.
- For tomatoes and sweet corn: one drip line per bed running along the centre.
- Emitter spacing must match the planned plant spacing for the crop as directed by the Agronomist.
- All drip lines must be connected to the header pipe and tested for flow before the mulch film is applied. Do not lay mulch film over an untested drip line.

2.4 Mulch Film Application

- Lay black mulch film over the bed surface after drip line installation and bed shaping.
- Pull the film taut along the full bed length before anchoring. A loose film collects water, tears easily, and loses its weed suppression function.
- Anchor the film edges by burying them in the soil on both sides of the bed or by pressing soil firmly over the edges at regular intervals.
- Cut planting holes using a sharp dibber, heated rod, or knife at the spacing specified by the Agronomist.
- Mulch film is not reused between crop cycles without Agronomist assessment and approval.

SECTION 3 - WATER SUPPLY AND IRRIGATION

3.1 System

The open field drip irrigation system is supplied from the same main water storage tank as the greenhouse. Water flows from the main tank through the delivery pipeline to the field header pipe and from there through the drip lines to each plant.

3.2 Irrigation Schedule

- Irrigation frequency and duration are set by the Agronomist based on crop stage, soil moisture, and weather conditions.
- In the absence of rainfall, open field crops typically require irrigation once or twice daily during establishment and active growth stages.
- During the rainy season, the Agronomist adjusts the schedule based on rainfall received. Do not irrigate if the beds have received adequate natural rainfall.
- The field worker checks soil moisture before every irrigation session. If the bed surface under the mulch film is still moist, report to the Agronomist before irrigating.

3.3 Pre-Irrigation Check

Before every irrigation run the field worker must:

- Walk the full length of every active bed and confirm all drip lines are intact and correctly positioned.
- Check all emitters along every bed. Clear or report any non-dripping emitter before the run begins.
- Confirm the water supply has adequate volume.
- Report all faults to the Agronomist before starting the run.

3.4 During Irrigation

- Walk every bed during every irrigation run. Do not leave the field during a run.
- Confirm every emitter is delivering water.
- Record the location of any pooling on the mulch film surface and report to the Agronomist.
- Record the location of any dry bed section adjacent to wet sections and address the emitter immediately.

3.5 After Irrigation

- Close all valves after the run is complete.
- Record the irrigation run on Form A: bed number, start time, end time, and any faults observed.

3.6 Rainy Season Management

- During active rainfall, switch off the irrigation system. Running drip irrigation during rain causes waterlogging and root disease.
- After heavy rainfall, inspect all beds for waterlogging and drainage failure. Report to the Agronomist immediately.
- Ensure inter-row walkways drain freely after rain. Standing water between beds must not persist for more than 24 hours.

SECTION 4 - DAILY ROUTINES

4.1 Morning Routine (7:00 AM)

4.1.1 Arrive at the field. Collect tools from the store.

4.1.2 Walk all three beds completely before any other activity. Conduct the full Morning Observation Walkthrough. See Section 5.

4.1.3 Check all drip lines and emitters on every bed. Report all faults to the Agronomist before irrigation begins.

4.1.4 Carry out the morning irrigation run as directed. Walk every bed during the run.

4.1.5 After irrigation, carry out the day's assigned tasks as directed by the Agronomist.

4.1.6 Complete morning entries on Form A.

4.2 Evening Routine (5:00 PM)

4.2.1 Walk all three beds completely. Conduct the Evening Observation Walkthrough. See Section 5.

4.2.2 Check all drip lines and emitters. Report new faults to the Agronomist.

4.2.3 Carry out the evening irrigation run if directed by the Agronomist.

4.2.4 Remove all plant debris, fallen fruits, and waste from bed surfaces and inter-row walkways.

4.2.5 Complete evening entries on Form A.

4.2.6 Return all tools to the store. Confirm the field area is secured before leaving.

SECTION 5 - OBSERVATION WALKTHROUGH

The observation walkthrough is conducted every morning and every evening on all three beds. Walk every row of every bed. Do not skip sections.

5.1 Overall Stand

Before entering the beds, observe the overall crop stand from the field edge. Note any areas where plants appear shorter, paler, or visibly stressed compared to surrounding plants.

5.2 Plant Body - Top to Bottom

Walk each bed and examine:

- Leaves for yellowing, browning, spots, curling, or distortion
- Stems for discolouration, softening, rot, or unusual narrowing
- Fruits for spots, cracks, sunken areas, premature drop, or abnormal colour
- Flowers for premature drop or failure to set

5.3 Leaf Undersides

Turn leaves over and examine at every session. Check for:

- Tiny insects, eggs, or crawling organisms
- Sticky residue indicating insect feeding activity
- Fine webbing between leaves or stems
- White powdery residue

5.4 Stem and Base

Examine the base of each plant at mulch film level. Check for:

- Dark soft rot or discolouration at the stem base
- Fungal growth appearing as white, grey, or brown fuzzy material
- Stem collapse or narrowing at soil level
- Widened planting holes exposing bare soil around the stem

5.5 Mulch Film and Bed Surface

Check along every bed for:

- Torn, lifted, or displaced mulch film
- Pooling water on the mulch film surface
- Dry sections of bed adjacent to wet sections
- Grass or weed growth pushing through or around mulch film edges

5.6 Drip Lines

Check for:

- Disconnected, kinked, or leaking drip lines
- Dry emitters after an irrigation run
- Lines displaced from their correct position

5.7 Inter-row Walkways

Check for:

- Weed growth between beds
- Standing water in the walkways
- Debris, waste, or abandoned tools

5.8 Record and Report

Record all observations on Form A. Use Section 6 to determine the appropriate response and timing.

SECTION 6 - OBSERVATION STANDARDS AND RESPONSE GUIDE

Table 6.1 - Open Field Observation Response Guide

What You See	Likely Cause	Action Required	When
Lower leaves yellowing across multiple plants	Nitrogen deficiency or waterlogging	Report to Agronomist. Record bed and row.	Same day
Interveinal chlorosis on older leaves	Magnesium or potassium deficiency	Report to Agronomist	Same day
Brown or black spots on leaves with yellow border	Fungal or bacterial infection	Do not remove leaves. Report to Agronomist immediately.	Immediately
White powder on leaf surface	Powdery mildew	Report to Agronomist immediately.	Immediately
Wilting despite recent irrigation	Root rot, stem rot, or severe pest damage	Check bed moisture. Report to Agronomist immediately.	Immediately
Distorted or curling new leaves	Aphid, mite, or virus	Check leaf undersides. Report to Agronomist.	Immediately
Sticky residue on leaves with visible insects	Aphid or whitefly infestation	Report to Agronomist. Record bed and rows.	Immediately
Fine webbing between leaves	Spider mite infestation	Report to Agronomist immediately.	Immediately
Holes in leaves or fruits with droppings visible	Caterpillar or armyworm	Report to Agronomist. Record bed and rows.	Immediately
Dark soft rot at stem base	Stem rot or Fusarium	Do not irrigate that plant. Report immediately.	Immediately
Fruit with sunken dark spots	Anthracnose or blossom end rot	Report to Agronomist. Do not harvest affected fruits.	Immediately
Sudden plant collapse	Bacterial wilt or stem borer	Remove plant immediately. Report to Agronomist.	Immediately
More than 10 percent of plants in a bed wilting	Widespread disease or irrigation failure	Report to Agronomist and Farm Manager immediately.	Immediately
Torn, lifted, or displaced mulch film	Physical damage or wind	Report to Agronomist. Record bed and location.	Same day
Dry bed section adjacent to wet sections	Blocked emitter or disconnected line	Clear or replace emitter. Report if unresolved.	Immediately
Pooling water on mulch film surface	Overwatering or drainage failure	Report to Agronomist.	Same day
Weed growth in inter-row walkways	Normal field condition	Remove by hand or hoe as part of daily routine.	Same day
Weed growth through mulch film edges	Film anchoring failure	Report to Agronomist. Re-anchor film as directed.	Same day

> ***Rule: The Field Worker observes and reports. No chemical application, plant removal, or system adjustment is made without the Agronomist's direct instruction.***

SECTION 7 - WEED MANAGEMENT

7.1 Mulch Film Coverage

The black mulch film covering the raised beds is the primary weed suppression tool. Maintaining the integrity of the mulch film is therefore a core weed management activity.

7.2 Inter-row Weed Control

- Weeds in the inter-row walkways are removed by hand or with a hoe during the morning routine whenever observed.
- Weeds must be removed before they flower or set seed. A weed that seeds in the field multiplies the problem for subsequent crops.
- Removed weeds are collected and taken away from the bed area immediately.
- Herbicides may be applied in the inter-row walkways only - never on or near the beds, mulch film, or plants. Any herbicide application requires Agronomist authorisation and Farm Manager awareness. Record on Form D.

7.3 Weed Growth Around Mulch Film Edges

- Weeds growing through or around the edges of the mulch film must be removed by hand immediately.
- Do not use a hoe near the mulch film edge as it risks tearing the film.
- If the mulch film edge has lifted and allowed weed establishment, re-anchor the film and report to the Agronomist.

7.4 Weekly Weed Assessment

Every Saturday the Agronomist conducts a full weed assessment of all three beds and inter-row walkways as part of the weekly inspection. The weed pressure level is recorded and a weeding schedule for the following week is assigned.

SECTION 8 - SWEET CORN OPERATIONS

8.1 Crop Overview

Sweet corn is a warm-season crop that grows rapidly from transplant to harvest. At WGNI Farm it is raised as seedlings in the nursery and transplanted to the open field beds. The crop cycle runs 70 to 90 days from transplanting to harvest depending on the variety.

8.2 Transplanting Standards

- Sweet corn seedlings are transplanted when they are 15 to 20 cm tall with 3 to 4 fully developed leaves.
- Plant spacing: 30 to 40 cm between plants within the row, 60 to 75 cm between rows.
- Transplant in the early morning or late afternoon. Never transplant during midday heat.
- Water each plant immediately after transplanting.
- Record the transplant date, bed number, and plant count on Form A.

8.3 Growth Stage Schedules

Transplanting Stage - Week 1

Daily Tasks

- Water every plant gently morning and evening.
- Walk every row. Report any wilting, fallen, or collapsed plant to the Agronomist immediately.
- Confirm every drip emitter is positioned correctly at each plant.
- Check mulch film holes for widening or lifting around new transplants.
- Remove any failed transplant and report to the Agronomist.
- Remove all debris and displaced soil from bed surfaces and walkways.

Weekly Tasks

- Record plant count per bed on Form A.
- Confirm drip lines are correctly routed along every row.
- Confirm mulch film edges are fully anchored.
- Full inter-row weed clearance.

Establishment Stage - Week 2 to Week 3

Daily Tasks

- Conduct full observation walkthrough morning and evening.
- Confirm every emitter is dripping and every plant is receiving water.
- Inspect the base of every stem for dark or soft rot. Report immediately.
- Remove all plant debris from bed surfaces and walkways.
- Report any plant still wilting 3 days after transplanting.

Weekly Tasks

- Check bed moisture at multiple points. The bed should be consistently moist but not waterlogged.
- Clean drip line filters.
- Full inter-row weed clearance.
- Record all plant losses on Form A.

Vegetative Stage - Week 3 to Week 6

Daily Tasks

- Conduct full observation walkthrough morning and evening.
- Check drip lines and emitters at every session.

- Inspect every plant for signs of pest activity particularly on leaf undersides. See Section 11.
- Remove all fallen leaves and debris from bed surfaces and walkways after every session.
- Report any plant growing significantly faster or slower than surrounding plants.

Weekly Tasks

- Agronomist assesses pest and disease pressure and directs control measures as required.
- Full inter-row weed clearance.
- Record plant population count and any losses on Form A.
- Flush drip lines at end of week.
- Full Saturday clean.

Tasselling and Silking Stage - Week 6 to Week 7

This is the pollination stage for sweet corn. The tassel (male flower at the top of the plant) sheds pollen which must reach the silks (female flower strands at the developing cob) for successful grain development. Poor pollination results in incomplete cob fill.

Daily Tasks

- Conduct full observation walkthrough morning and evening.
- Observe tassels on every plant. A tassel that is brown and dry without having shed pollen indicates stress or disease. Report to the Agronomist.
- Observe silks on developing cobs. Silks should be cream to light brown during pollination. Dark brown shrivelled silks before the cob has developed indicate pollination failure. Report to the Agronomist.
- Do not apply any chemical spray during the tasselling and silking period. Chemical contact with pollen and silks reduces pollination success.
- Check drip lines and emitters at every session. Water stress during this stage causes severe yield loss.
- Remove all debris from bed surfaces and walkways.

Weekly Tasks

- Agronomist assesses pollination progress and cob development.
- Record observations on Form A including tassel and silk condition across all beds.
- Full Saturday clean.

Tasselling and Silking Stage Standards

- Temperature during this stage must not exceed 35 degrees Celsius for extended periods. Pollen viability drops sharply above this temperature. Ensure full irrigation is maintained during heat events.
- Wind assists natural pollen dispersal. Do not erect barriers that block airflow around the crop during this stage.

Grain Fill Stage - Week 7 to Week 9

Daily Tasks

- Conduct full observation walkthrough morning and evening.
- Inspect developing cobs. Report any cobs with abnormal shape, discolouration, or pest damage to the Agronomist.
- Check drip lines and emitters at every session. Consistent moisture is critical during grain fill.
- Remove all debris from bed surfaces and walkways.

Weekly Tasks

- Agronomist assesses cob development and estimates harvest readiness.
- Record estimated cob count per bed on Form A.
- Full Saturday clean.

8.4 Harvest Guide - Sweet Corn

Sweet corn has a narrow harvest window. Over-mature corn loses sweetness rapidly.

Table 8.1 - Sweet Corn Harvest Readiness Indicators

Indicator	Standard
Silk condition	Silks have turned completely brown and dried
Cob feel	Well-filled and firm when pressed along its length
Top of husk	Feels solid when pressed - no air gap between husk and grain
Kernel test	A punctured kernel releases milky liquid. Clear liquid means under-ripe. Thick paste means over-ripe.
Days from silk	Approximately 18 to 24 days after silks first appear - variety dependent

Harvesting Procedure:

- The Agronomist confirms harvest readiness before picking begins.
- Hold the stalk firmly with one hand and the cob with the other. Twist and pull the cob downward in one firm motion.
- Place harvested cobs immediately into clean crates. Do not place on bare ground.
- Harvest in the early morning when temperatures are lowest. Sweet corn loses sweetness faster in heat.
- Record the harvest quantity per bed on Form E.

Post-Harvest:

- Move harvested cobs to the designated post-harvest area immediately. Do not leave in the sun or in the field after picking.
- Grade cobs: Grade A (full, well-developed, no pest damage), Grade B (minor imperfection, marketable), Reject (damaged, diseased, underdeveloped).

8.5 End of Cycle

- Remove all stalks and crop residue from the bed when harvest is complete.
- Do not leave crop residue in the bed. It harbours pests and disease for the next crop.
- The Agronomist assesses the bed and directs the next crop cycle preparation.

SECTION 9 - TOMATO OPERATIONS

9.1 Crop Overview

Tomato is a warm-season fruiting crop with a longer production cycle than sweet corn. At WGNI Farm, tomatoes are raised as seedlings in the nursery and transplanted to the open field beds. The crop cycle runs approximately 90 to 120 days from transplanting to end of harvest depending on variety, with continuous harvesting over several weeks once fruiting begins.

9.2 Transplanting Standards

- Tomato seedlings are transplanted when they are 15 to 20 cm tall, have 4 to 6 true leaves, and show no signs of pest or disease.
- Plant spacing: 45 to 60 cm between plants within the row, 75 to 90 cm between rows.
- Transplant in the early morning or late afternoon. Never transplant during midday heat.
- Water each plant immediately after transplanting.
- Apply a preventive drench of Metalaxyl or Thiram around the base of each transplant at planting to protect against damping off and early soil-borne disease.
- Record the transplant date, bed number, variety, and plant count on Form A.

9.3 Growth Stage Schedules

Transplanting Stage - Week 1

Daily Tasks

- Water every plant gently morning and evening.
- Walk every row. Report any wilting, fallen, or collapsed plant immediately.
- Confirm drip emitter is correctly positioned at each plant.
- Check mulch film holes for widening or lifting.
- Remove any failed transplant and report to the Agronomist.
- Remove all debris from bed surfaces and walkways.

Weekly Tasks

- Record plant count per bed.
- Confirm drip lines routed correctly.
- Confirm mulch film edges anchored.
- Full inter-row weed clearance.

Establishment Stage - Week 2 to Week 3

Daily Tasks

- Conduct full observation walkthrough morning and evening.
- Confirm every emitter is dripping and every plant is receiving water.
- Inspect the base of every stem for dark or soft rot. Report immediately.
- Remove all plant debris from beds and walkways.
- Report any plant still wilting 3 days after transplanting.

Weekly Tasks

- Check bed moisture at multiple points.
- Clean drip line filters.
- Full inter-row weed clearance.
- Record all plant losses on Form A.

Vegetative Stage - Week 3 to Week 6

Daily Tasks

- Conduct full observation walkthrough morning and evening.
- Check drip lines and emitters at every session.
- Inspect every plant for signs of pest activity particularly on leaf undersides. See Section 11.
- Remove all fallen leaves and debris from beds and walkways.
- Check and maintain stakes and ties on all plants. See Section 9.5.
- Remove suckers as directed by the Agronomist.
- Report any plant showing rapid wilting, stem discolouration, or unusual growth.

Weekly Tasks

- Agronomist conducts pruning and sucker removal assessment.
- Full inter-row weed clearance.
- Record plant population count and losses.
- Flush drip lines.
- Full Saturday clean.

Flowering Stage - Week 6 to Week 8

Daily Tasks

- Conduct full observation walkthrough morning and evening.
- Monitor for flower drop. If more than 20 percent of flowers are dropping without setting fruit across a bed, report to the Agronomist same day.
- Check drip lines and emitters at every session.
- Do not apply chemical sprays during peak flowering hours. Apply in early morning before 9:00 AM or late afternoon after 4:00 PM.
- Remove all debris from beds and walkways.

Weekly Tasks

- Agronomist assesses fruit set rate and adjusts nutrient programme.
- Increase calcium and potassium in the fertigation programme at this stage.
- Check and tighten all staking and ties.
- Full Saturday clean.
- Record all flower drop observations on Form A with bed number and date.

Flowering Stage Standards

- Temperature must remain below 35 degrees Celsius during flowering. Temperatures above this cause widespread flower drop and poor fruit set.
- Humidity above 85 percent encourages early blight and Botrytis. Ensure good air movement around the plants.
- Never withhold water during the flowering stage. Water stress is a primary cause of blossom end rot.

Fruit Development Stage - Week 8 to Week 12

Daily Tasks

- Conduct full observation walkthrough morning and evening.
- Inspect every fruit. Report spots, cracks, sunken areas, blossom end rot, or abnormal colour to the Agronomist immediately.
- Check drip lines and emitters at every session.
- Check and reinforce stakes and ties. Fruit load increases significantly at this stage.
- Remove all debris from beds and walkways.

Weekly Tasks

- Agronomist assesses fruit development against expected size benchmarks.
- Adjust irrigation to maintain consistent moisture. Irregular watering causes blossom end rot and fruit cracking.

- Record estimated fruit count per bed.
- Full Saturday clean.

Harvest Stage - Week 12 Onwards

Daily Tasks

- Assess fruits for harvest readiness. See Section 9.4.
- Harvest ripe fruits every 2 to 3 days.
- After harvest, inspect for diseased or damaged fruits remaining on the plant. Remove and present to the Agronomist before disposal.
- Check drip lines and emitters at every session.
- Remove all fallen or damaged fruits from beds and walkways immediately.

Weekly Tasks

- Record total harvest by weight and grade on Form E.
- Assess overall plant condition and vigour. Report any significant decline.
- Full Saturday clean.
- Agronomist reviews crop performance and advises on continuation or cycle termination.

9.4 Harvest Guide - Tomatoes

Table 9.1 - Tomato Harvest Readiness Standards

Indicator	Standard
Colour	Fruit has reached the expected colour for the variety - red, pink, or yellow fully developed
Firmness	Firm but with a slight give when pressed gently. Not hard or rock-solid.
Size	Full size for the variety reached
Stem attachment	Fruit detaches easily from the stem with a gentle twist
Days from transplant	Approximately 80 to 100 days depending on variety

Harvesting Procedure:

- The Agronomist confirms harvest readiness before picking begins.
- Use clean hands or clean harvest scissors. Twist the fruit gently or cut the stem cleanly leaving a short stub attached to the fruit.
- Handle fruits carefully. Do not drop, compress, or stack roughly.
- Place harvested fruits immediately into clean food-grade harvest crates. Do not place on bare ground.
- Harvest in the early morning when temperatures are lowest to maximise shelf life.
- Record the harvest quantity per bed on Form E.

Grading:

- Grade A: Full size, correct colour, no blemish, firm.
- Grade B: Minor surface imperfection, slightly irregular shape, marketable.
- Reject: Cracked, soft, diseased, insect-damaged, or significantly discoloured.

9.5 Staking and Training - Tomatoes

Tomatoes must be staked and trained to grow upright. Without support, plants collapse under fruit weight, increasing disease pressure and reducing yield quality.

Stakes

- Drive a stake firmly into the ground 5 to 10 cm from each plant immediately after transplanting.

- Stake height: minimum 1.2 to 1.5 metres above ground.
- Use bamboo stakes, wooden stakes, or any firm material designated by the Agronomist.

Tying

- Tie the main stem to the stake using soft twine. The loop must be loose - two fingers must fit between the twine and the stem.
- Add a new tie every 20 to 25 cm of new growth.
- Never tie tightly against the stem. A tight tie damages the vascular tissue and causes stem narrowing and death above the tie point.
- Check all ties weekly. Replace any that are cutting into the stem or have broken.

Sucker Removal

- Suckers are lateral shoots growing from the junction between the main stem and a leaf stalk. They divert energy from fruit production.
- Suckers are removed as directed by the Agronomist, typically during the vegetative stage.
- Remove suckers by pinching them off with clean fingers when they are small - under 5 cm. Larger suckers are removed with clean disinfected scissors.
- Disinfect scissors with Hypo solution between plants when working in any area with disease pressure.

9.6 End of Cycle

- Remove all plants, stakes, ties, and crop residue from the bed when harvest is complete.
- Do not leave crop residue in or near the bed. It harbours pests and disease for the next crop.
- The Agronomist assesses the bed and directs the next crop cycle preparation.

SECTION 10 - NUTRIENT MANAGEMENT

10.1 Basal Fertiliser Application

Before bed formation, basal fertiliser is incorporated into the soil as directed by the Agronomist. Commonly used basal fertilisers in Nigerian open field vegetable production: NPK 15:15:15, NPK 20:10:10, or as directed by the Agronomist based on soil assessment. Application rate, timing, and method are determined solely by the Agronomist.

10.2 Fertigation Programme

Table 10.1 - Nutrient Programme by Growth Stage

Growth Stage	Nutrient Focus	Notes
Establishment (Week 1 to 2)	Phosphorus dominant	Promotes root development. Keep nitrogen low at this stage.
Vegetative (Week 3 to 6)	Nitrogen dominant	Drives leaf and stem growth.
Flowering (Week 6 to 8)	Increase Potassium and Calcium	Reduces nitrogen. Supports flower development. Prevents blossom end rot.
Fruit Development (Week 8 to 12)	Potassium dominant, sustained Calcium	Supports fruit size, colour, and shelf life.
Harvest Period	Balanced - maintain K and Ca	Do not reduce nutrients during active harvest.

10.3 Foliar Nutrition

Foliar feeding supplements the fertigation programme. It is applied as a fine spray directly onto the leaf surface and is used to correct visible deficiencies rapidly or to support the crop during high-demand stages. All foliar applications are authorised and supervised by the Agronomist only.

Table 10.2 - Common Foliar Nutrient Formulations

Formulation	Nutrient Profile	Primary Use
Balanced NPK (e.g. 20:20:20 or similar)	Equal N, P, K	General growth support during vegetative stage
High-Potassium soluble (e.g. 9:18:36 or similar)	Low N, high K	Flowering and fruit development
Calcium-Magnesium supplement	Ca and Mg	Prevents blossom end rot and interveinal chlorosis
Iron chelate (Fe-EDTA)	Chelated Iron	Correction of iron deficiency on young leaves
Boron supplement	Boron (B)	Pre-flowering - supports pollen viability and fruit set

Application Rules:

- Apply in the early morning before 9:00 AM or late afternoon after 4:00 PM only. Never apply during midday heat.
- Do not apply foliar feed and pesticide on the same day unless the Agronomist confirms compatibility.
- Do not apply foliar feed during rainfall or when rain is expected within 2 hours.
- Wear gloves, nose mask, and long-sleeved clothing during all foliar applications.
- Record every foliar application on Form D immediately.

10.4 Nutrient Deficiency Guide

Table 10.3 - Nutrient Deficiency Visual Guide

Nutrient	Symptom	First Appears On	Action
Nitrogen (N)	General yellowing, pale appearance, slow growth	Older lower leaves	Report to Agronomist same day
Phosphorus (P)	Purple or reddish colour on leaf underside	Older leaves	Report to Agronomist same day
Potassium (K)	Brown scorching at leaf edges and tips	Older leaves	Report to Agronomist same day
Calcium (Ca)	Blossom end rot on fruits. Tip burn on young leaves.	Young leaves and fruits	Report to Agronomist immediately
Magnesium (Mg)	Interveinal chlorosis - yellowing between veins, veins stay green	Older leaves	Report to Agronomist same day
Iron (Fe)	Interveinal chlorosis on young leaves only	Young new leaves	Report to Agronomist same day
Boron (B)	Distorted new growth, hollow fruit interior	Young growth and fruits	Report to Agronomist immediately

SECTION 11 - PEST AND DISEASE CONTROL

11.1 Operational Rules

- No chemical may be applied without the Agronomist's direct authorisation and the Farm Manager's awareness.
- Every chemical application must be recorded on Form D immediately after completion.
- Full protective equipment must be worn: gloves, nose mask, protective eyewear, and long-sleeved clothing.
- Never spray chemicals on windy days. Spray drift damages neighbouring plants and reduces efficacy.
- Observe the re-entry interval stated on every product label before re-entering treated areas.

11.2 Pest Identification and Chemical Control

Table 11.1 - Common Open Field Pests and Recommended Treatments

Pest	Crops Affected	Symptoms	Recommended Chemical	Application	Frequency
Aphids	Tomato, Corn	Clusters of soft insects on new growth. Sticky film. Curling leaves.	Imidacloprid, Dimethoate, Cypermethrin	Foliar - new growth and leaf undersides	Every 7 days until controlled
Whitefly	Tomato	White insects fly up when plant disturbed. Sticky film. Leaf yellowing.	Cypermethrin, Imidacloprid, Thiamethoxam (Actara)	Foliar - leaf undersides	Every 7 days. Rotate chemicals.
Spider Mites	Tomato	Fine webbing. Bronze or silver leaf discolouration.	Abamectin (Dynamec), Dicofol, Kelthane	Foliar - both leaf surfaces	Every 5 to 7 days. Rotate.
Thrips	Tomato, Corn	Silver streaks on leaves and fruits. Tiny insects in flowers.	Spinosad (Tracer), Imidacloprid, Abamectin	Foliar - leaves and flower area	Every 7 days. Rotate.
Armyworm and Caterpillars	Corn, Tomato	Holes in leaves, tassels, cobs or fruits. Droppings visible.	Cypermethrin, Lambda-cyhalothrin (Karate), Chlorpyrifos, Emamectin benzoate (Proclaim)	Foliar spray	When observed. Early morning preferred.
Fall Armyworm	Corn	Holes in leaves with frass. Damage to tassel and cob. Larvae in whorl.	Emamectin benzoate (Proclaim), Chlorpyrifos, Lambda-cyhalothrin (Karate)	Directed spray into whorl	Every 5 to 7 days during infestation
Stalk Borer	Corn	Entry hole at stem base. Plant wilts from growing point. Dead heart symptom.	Carbofuran (Furadan) at planting, Lambda-cyhalothrin (Karate) foliar	Soil at planting then foliar	Preventive at planting. Foliar when observed.

Pest	Crops Affected	Symptoms	Recommended Chemical	Application	Frequency
Leaf Miners	Tomato	Winding pale tunnels inside leaf tissue when held to light.	Abamectin (Dynamec), Cyromazine (Trigard)	Foliar spray	Every 7 to 10 days
Root Knot Nematode	Tomato	Stunted wilting plant unresponsive to irrigation. Root galls visible.	Carbofuran (Furadan)	Soil drench into bed	Preventive at bed preparation

11.3 Disease Identification and Chemical Control

Table 11.2 - Common Open Field Diseases and Recommended Treatments

Disease	Crops Affected	Symptoms	Recommended Chemical	Application	Frequency
Early Blight	Tomato	Circular dark spots with concentric rings on older leaves. Yellowing around spots.	Mancozeb (Dithane M-45), Chlorothalonil (Bravo), Carbendazim	Foliar spray	Every 7 to 10 days. Begin preventively.
Late Blight	Tomato	Water-soaked lesions on leaves rapidly turning brown. White mould on leaf underside in humid conditions. Spreads very fast.	Metalaxyl (Ridomil), Cymoxanil plus Mancozeb (Curzate M), Dimethomorph	Foliar spray	Every 5 to 7 days. Increase during wet weather.
Fusarium Wilt	Tomato	One-sided yellowing and wilting. Brown vascular discolouration when stem is cut.	No curative. Remove plant. Drench surrounding soil with Carbendazim.	Soil drench	Immediately on detection
Bacterial Wilt	Tomato	Sudden total wilting. Brown vascular tissue visible when stem is cut.	No curative. Remove plant. Drench surrounding soil with Copper Oxychloride.	Soil drench	Immediately on detection
Blossom End Rot	Tomato	Dark sunken patch at the blossom end of the fruit. Calcium deficiency aggravated by irregular watering.	Calcium foliar spray. Regulate irrigation.	Foliar and irrigation management	When observed

Disease	Crops Affected	Symptoms	Recommended Chemical	Application	Frequency
Anthracnose	Tomato	Sunken dark circular spots on ripe and ripening fruits.	Mancozeb, Chlorothalonil (Bravo), Carbendazim	Foliar and fruit spray	Every 7 to 10 days
Damping Off	Tomato (seedlings)	Seedlings collapse at soil level. Stem rotted at base.	Metalaxyl, Thiram, Captan	Growing medium drench	Preventive at transplanting
Grey Leaf Spot	Corn	Small circular spots with grey centres and dark borders on leaves.	Mancozeb, Carbendazim	Foliar spray	Every 7 to 10 days
Northern Corn Leaf Blight	Corn	Large elongated tan-coloured lesions on leaves. 2 to 15 cm long.	Mancozeb, Propiconazole (Tilt)	Foliar spray	Every 7 to 10 days when observed
Botrytis Grey Mould	Tomato	Grey fuzzy mould on stems, leaves, and fruits. Common in cool humid conditions.	Iprodione (Rovral), Carbendazim, Thiram	Foliar spray	Every 7 days during humid periods

11.4 Chemical Rotation

No chemical active ingredient may be applied more than twice consecutively for the same pest or disease. On the third application, a different active ingredient must be used. The Agronomist manages the rotation schedule and records every application on Form D.

11.5 Chemical Safety Rules

- Wear full protective gear during every application: gloves, nose mask, protective eyewear, long-sleeved clothing.
- Never spray against the wind direction.
- Do not eat, drink, or smoke during or immediately after application.
- After application, wash all exposed skin and hands thoroughly with soap and water.
- Return all chemicals to the locked chemical store immediately after use.
- Empty containers are disposed of safely. Never burn or reuse chemical containers.

SECTION 12 - CLEANING AND FIELD SANITATION

12.1 Daily Sanitation

- Remove all fallen leaves, crop debris, and damaged fruits from bed surfaces and inter-row walkways after every session.
- Any plant material removed from a diseased plant must be placed in a sealed bag and removed from the field immediately. Do not compost diseased material.
- Tools used on diseased plants must be disinfected with Hypo solution (1 part Hypo to 9 parts water) before use on healthy plants.
- Return all tools to the store at the end of every session.

12.2 Weekly Cleaning - Every Saturday

The full weekly field sanitation is carried out every Saturday. The Agronomist inspects after cleaning and signs Form C.

Weekly tasks:

- Full removal of all plant debris from bed surfaces, inter-row walkways, and the field perimeter.
- Full inter-row weed clearance across all three beds.
- Inspect and clean drip line filters. Flush all lines with clean water.
- Inspect all mulch film across all beds. Record and report all tears, lifting, or displacement.
- Clean and disinfect all field tools using Hypo solution.
- Inspect the field perimeter for encroachment, boundary damage, or security concerns. Report to Farm Manager.

12.3 End of Crop Cycle Cleanout

At the end of every crop cycle:

- Remove all plant material including stems, roots, and any remaining fruits from the bed entirely.
- Remove all mulch film from the bed. Dispose off the farm or as directed by the Agronomist. Do not burn mulch film on the field.
- Turn and till the bed soil to a depth of at least 20 cm. Expose soil to sun for a minimum of 7 days before the next crop.
- Apply lime to the bed soil if directed by the Agronomist to adjust pH and reduce pathogen load.
- Install fresh mulch film before the next transplanting cycle.
- Record end of cycle cleanout on Form A with date and bed number.

SECTION 13 - ORCHARD OPERATIONS

13.1 Overview

The WGNI Farm orchard is a designated area for the establishment and long-term management of fruit trees including mango and orange, with additional species to be introduced as the farm expands. The orchard is managed as a permanent growing infrastructure that will produce fruit over many seasons. Every decision made during establishment directly affects the productivity and longevity of the orchard.

13.2 Site Demarcation and Preparation

13.2.1 Full Demarcation Standard

The orchard area must be fully and clearly demarcated before any tree planting begins.

- The full orchard boundary must be marked with permanent stakes or fencing at all corners and at regular intervals along each boundary.
- No livestock, vehicles, or unauthorised personnel may access the demarcated orchard area.
- A 2-metre buffer strip around the inside of the orchard boundary must be kept clear of weeds, debris, and any material that could harbour pests or rodents.

13.2.2 Land Clearing

- Remove all existing vegetation from the orchard site including grasses, weeds, shrubs, and tree stumps.
- Remove all roots from the soil to prevent regrowth and to allow proper land levelling.
- Level the site to eliminate waterlogging risks. Fruit trees do not tolerate waterlogged roots.
- The Agronomist assesses the cleared site before any planting position is marked.

13.2.3 Tree Spacing

Table 13.1 - Recommended Tree Spacing

Tree Species	Minimum Spacing (Tree to Tree)	Minimum Spacing (Row to Row)
Mango	8 to 10 metres	8 to 10 metres
Orange (Citrus)	5 to 6 metres	5 to 6 metres
Other species	As determined by Agronomist	As determined by Agronomist

- Tree positions are marked on the ground with stakes before any hole is dug.
- The Agronomist approves the tree layout before hole digging begins.

13.2.4 Planting Hole Preparation

- Planting holes are dug to a minimum depth of 60 cm and width of 60 cm for all fruit tree species.
- The topsoil removed from the hole is kept separate from the subsoil.
- Each hole is filled with a prepared mix of topsoil, compost or well-rotted manure, and any basal fertiliser specified by the Agronomist.
- Holes are prepared and filled a minimum of 2 weeks before the seedlings arrive to allow the soil mix to settle.

13.3 Seedling Sourcing and Standards

Fruit tree seedlings have not yet been sourced at the time of this SOP. The following standards must be applied when sourcing begins:

- Seedlings must be sourced from a reputable and certified nursery only. The Agronomist and Farm Manager approve the source before procurement.
- Seedlings must be grafted onto a disease-resistant rootstock where possible.

On arrival, all seedlings must be inspected before accepting delivery. Reject any seedling showing:

- Root rot or root damage
- Pest infestation including scale insects, mealybugs, or aphid colonies
- Disease symptoms including leaf spots, stem lesions, or unusual discolouration

- Poor root development
- Damaged or broken stems
- Do not accept seedlings that have been in transit for more than 48 hours without adequate water.
- Record all seedling delivery details on Form A: species, quantity, source nursery, date of arrival, and inspection result.

13.4 Tree Planting

13.4.1 Planting Procedure

1. Water the seedling thoroughly in its container at least 1 hour before planting.
2. Remove the seedling from its container carefully. Do not disturb the root ball.
3. Place the root ball in the centre of the prepared planting hole. The graft union must sit at least 10 cm above the final soil level. Never bury the graft union.
4. Fill in around the root ball with the prepared soil mix. Firm gently as you fill to eliminate air pockets around the roots.
5. Form a shallow watering basin around each tree by creating a low soil ring approximately 30 cm from the stem.
6. Water immediately and thoroughly after planting.
7. Apply mulch around the base of each tree. See Section 13.5.
8. Install a support stake beside each tree immediately after planting.

13.4.2 Planting Timing

- Plant at the onset of the rainy season where possible. Natural rainfall supports establishment and reduces irrigation demand during the critical first weeks.
- If planting in the dry season, ensure reliable irrigation is in place before planting begins.
- Plant in the early morning or late afternoon. Never plant during midday heat.

13.5 Tree Establishment - Year 1

13.5.1 Mulching

- Apply a layer of organic mulch (dry grass, straw, wood chips, or sawdust) around the base of each newly planted tree. Mulch depth: 5 to 10 cm.
- The mulch must not touch the tree stem. Leave a clear gap of at least 10 cm between the mulch and the stem to prevent stem rot.
- Mulch radius: extend the mulch to cover the full base circle of at least 60 cm radius around the tree.
- Replace and replenish mulch as it decomposes throughout the year.

13.5.2 Staking

- Drive a firm stake into the ground beside each newly planted tree within 10 cm of the stem.
- Tie the tree to the stake using soft twine. The tie must be loose - two fingers must fit between the twine and the stem.
- Check all ties monthly. Replace any that are cutting into the bark.
- Stakes may be removed after the tree has established firmly - typically after 12 to 18 months. The Agronomist confirms when stakes are no longer needed.

13.5.3 Irrigation During Establishment

- During the first 3 months: water every 2 to 3 days if there is no significant rainfall.
- After 3 months: reduce to twice weekly during the dry season.
- During the rainy season: only irrigate if there has been no significant rainfall for 5 or more days.
- Never allow newly planted trees to reach the point of severe wilting. Establishment-stage stress causes permanent root damage and stunted tree development.

13.6 Orchard Ground Management

13.6.1 Base Circle

- The area within 60 cm of each tree trunk must be kept free of grass and weeds at all times.
- Remove grass and weeds from the base circle by hand only. Herbicides must not be applied within 60 cm of the tree trunk.
- Organic mulch is maintained within the base circle at all times.

13.6.2 Between Tree Rows

- Grass between tree rows is managed by cutting short - not eliminating entirely. A short grass cover between rows protects soil structure, reduces erosion, and provides a pathway surface.
- Grass between rows must not exceed 15 cm in height at any time.
- Cutting is carried out using a lawn mower, slasher, or brush cutter as directed by the Farm Manager.
- Aggressive or invasive weed species in the inter-row must be removed at the root rather than simply slashed.

13.6.3 Herbicide Use in the Orchard

- Herbicides may be applied to the inter-row areas only - never within the base circle of any tree.
- All herbicide applications require Agronomist authorisation and Farm Manager awareness.
- Contact herbicides (e.g. Paraquat - Gramoxone) may be used for inter-row weed control. They must never be allowed to contact the tree foliage, stem, or root zone.
- Record every herbicide application on Form D.

13.7 Orchard Maintenance - Ongoing

Daily Tasks During Establishment Year

- Walk the full orchard every morning. Observe every tree.
- Check for wilting, stem damage, pest activity, or unusual discolouration on any tree.
- Check that all stakes and ties are intact.
- Check the base circle of each tree for weed encroachment.
- Report any abnormal observation to the Agronomist immediately.

Weekly Tasks

- Hand-weed the base circle of every tree.
- Check and replenish mulch where it has thinned or decomposed.
- Inspect tree ties. Replace any that are tight or damaged.
- Slash inter-row grass if height exceeds 15 cm.
- Record orchard maintenance activities on Form F.
- Agronomist inspects all trees and records observations on Notion.

13.7.3 Pruning - Formative Pruning (Year 1 to Year 3)

Formative pruning shapes the young tree into a productive and well-structured canopy. It is carried out exclusively by the Agronomist during the first three years.

- Remove any crossing branches that rub against each other.
- Remove any downward-growing branches.
- Remove any branch growing toward the centre of the tree that will block light and air circulation as the tree matures.
- For mango: maintain 3 to 4 main scaffold branches from the central leader. Remove any competing leaders.
- For orange: maintain an open-centre or modified leader structure as directed by the Agronomist.
- All pruning cuts must be clean using sharp, disinfected pruning tools.
- Apply wound sealant (tar-based sealant or copper-based paste) to any pruning cut larger than 2 cm in diameter to prevent disease entry.
- Record all pruning events on Form F.

13.8 Pest and Disease Management - Orchard

Table 13.2 - Common Orchard Pests and Recommended Treatments

Pest	Trees Affected	Symptoms	Recommended Chemical	Application	Frequency
Mango Mealybug	Mango	White waxy clusters on stems, leaves, and fruit clusters. Honeydew and sooty mould.	Imidacloprid, Chlorpyrifos, Dimethoate	Foliar spray - stem and leaf	Every 14 days during active infestation
Scale Insects	Mango, Orange	Small brown or white oval shells on stems and leaves. Yellowing and dieback.	White oil (mineral oil), Chlorpyrifos, Dimethoate	Foliar spray	Every 14 days until controlled
Mango Seed Weevil	Mango	Damage visible inside the seed after harvest. No external symptoms on fruit.	Dimethoate foliar spray during flowering	Foliar during flowering	As directed by Agronomist
Aphids	Orange	Colonies on new growth. Curling leaves. Sticky honeydew.	Imidacloprid, Dimethoate, Cypermethrin	Foliar - new growth	Every 7 to 14 days until controlled
Citrus Psyllid	Orange	Distorted new growth. Yellowing shoots. Tiny insects and waxy deposits visible.	Imidacloprid, Dimethoate	Foliar spray	Every 7 to 14 days during flush growth
Fruit Fly	Mango, Orange	Puncture marks on fruit surface. Maggots inside ripe fruit. Premature fruit drop.	Protein bait stations (Malathion bait), Dimethoate	Bait traps and spot spray	Weekly during fruiting season

Table 13.3 - Common Orchard Diseases and Recommended Treatments

Disease	Trees Affected	Symptoms	Recommended Chemical	Application	Frequency
Mango Anthracnose	Mango	Black spots on leaves, flowers, and fruits. Blossom blight. Dark sunken spots on ripe fruit.	Mancozeb, Copper Oxychloride, Carbendazim	Foliar spray	Every 7 to 14 days especially during flowering and wet weather
Powdery Mildew	Mango	White powdery coating on young leaves, flowers, and fruits. Causes flower and fruit drop.	Sulfur fungicide (Thiovit), Triadimefon (Bayleton), Myclobutanil	Foliar spray	Every 7 to 10 days during flowering
Citrus Gummosis	Orange	Gum oozing from bark on the lower trunk. Bark	Metalaxyl (Ridomil), Fosetyl-aluminium	Soil drench and bark application	When observed. Improve drainage

Disease	Trees Affected	Symptoms	Recommended Chemical	Application	Frequency
		becomes dark and dies.	(Aliette) - drench and paint on affected bark		around tree.
Citrus Canker	Orange	Raised corky lesions on leaves, stems, and fruits with yellow halo.	Copper Oxychloride, Copper hydroxide	Foliar spray	Every 14 days as preventive
Sooty Mould	Mango, Orange	Black sooty coating on leaves and stems - secondary to honeydew from insects.	Control the insect causing honeydew first. Clean leaves with dilute Hypo spray.	Foliar - insect control primary	Treat underlying pest

SECTION 14 - FARM MANAGER OVERSIGHT

14.1 Daily Responsibilities

- Collect Form A from the Agronomist every evening. Review all entries.
- Verify that Notion records have been updated before 7:00 PM daily.
- Follow up on all flagged observations, pest or disease reports, irrigation faults, or missed records before the following morning.
- Confirm the field area and orchard are secure at the end of each working day.

14.2 Weekly Inspection - Every Saturday

The Farm Manager conducts a formal inspection of all three seed beds and the orchard every Saturday alongside the Agronomist.

Table 14.1 - Weekly Open Field and Orchard Inspection Checklist

Area	What to Check	Acceptable Standard
Plant health - all three beds	Colour, uniformity, vigour	No widespread yellowing, wilting, or unmanaged pest or disease
Drip lines and emitters - all beds	All functioning, no leaks	Every emitter delivering. No pooling or dry bed sections.
Mulch film - all beds	Condition and integrity	No tears, lifting, or displaced sections
Inter-row walkways - all beds	Weed level and drainage	Weeds below 10 cm. No standing water.
Staking and ties - tomatoes	All plants supported	No collapsed or unsupported plants. No tight ties.
Chemical store	Labelled, locked, organised	No unlabelled containers. Store secured.
Field perimeter	Boundary integrity	No encroachment, gaps, or security concerns
Orchard demarcation	Boundary markers intact	All stakes or fencing visible and in position
Orchard trees	Health, stakes, ties, mulch	No wilting, pest activity, or base circle weed encroachment
Orchard base circles	Weed-free and mulched	Clear of grass and weeds. Mulch maintained.
Inter-row grass - orchard	Grass height	Below 15 cm at all times
Form A completeness	All entries filled and signed	No missing entries or unsigned days
Form F completeness	Orchard log entries current	All weekly orchard activities recorded

14.3

The Farm Manager signs Form C every Saturday after the inspection is complete.

14.4

The Farm Manager approves all chemical, fertiliser, and seedling purchases before procurement. No input enters the farm without Farm Manager authorisation.

14.5

The Farm Manager does not override Agronomist crop or orchard decisions without consultation. Unresolved disagreements are escalated to the CEO.

SECTION 15 - NOTION RECORDS

15.1

The Agronomist updates the farm Notion system daily before 7:00 PM with the following for each active bed and the orchard:

- Irrigation run details
- Pest or disease observations and actions taken
- Plant and tree losses
- Chemical and foliar applications with product name, rate, and area treated
- Nutrient programme adjustments
- Harvest records
- Orchard observations and maintenance activities
- Any escalations to the Farm Manager

15.2

Physical forms (Form A through Form F) are the field record of all daily operations and must be completed in full before Notion entry. Both must be maintained without exception.

15.3

The Farm Manager reviews Notion records daily and confirms completeness. Any gap in the Notion record is treated as a compliance failure.

SECTION 16 - SANCTION GRID

Table 16.1 - Disciplinary Sanction Grid

Breach	First Offence	Second Offence	Third Offence
Applying any chemical without Agronomist authorisation	Suspension.	Termination.	N/A
Applying foliar feed without Agronomist instruction	Verbal warning. Recorded.	Written warning.	Suspension.
Leaving crop debris in the field after session	Verbal warning. Recorded.	Written warning.	Pay deduction.
Failing to report a pest or disease observation same day	Written warning.	Suspension.	Termination.
Harvesting without Agronomist confirmation	Written warning.	Suspension.	Termination.
Tying plant or tree ties tightly against stem or bark	Verbal warning. Recorded.	Written warning.	Suspension.
Allowing tools to remain in the field overnight	Verbal warning. Recorded.	Written warning.	Pay deduction.
Leaving mulch film damage unreported same day	Verbal warning. Recorded.	Written warning.	Pay deduction.
Allowing inter-row weeds to flower or seed without reporting	Written warning.	Written warning.	Pay deduction.
Using herbicide within the orchard base circle	Written warning.	Suspension.	Termination.
Allowing livestock or vehicles into the orchard	Written warning.	Suspension.	Termination.
Form A or Form F not completed or submitted same day	Verbal warning. Recorded.	Written warning.	Pay deduction for the day.
Removing plant or tree material without Agronomist instruction	Verbal warning. Recorded.	Written warning.	Suspension.
Allowing newly planted trees to wilt to severe stress	Written warning.	Suspension.	N/A
Chemical containers not returned to locked store after use	Written warning.	Suspension.	Termination.
Agronomist failing to update Notion records before 7:00 PM	Verbal warning. Recorded.	Written warning.	Escalation to Farm Manager.
Farm Manager failing to conduct Saturday weekly inspection	Verbal warning from CEO. Recorded.	Written report to CEO.	N/A

FORMS AND RECORDS

FORM A - WEEKLY DAILY ENTRY FORM

WGNI FARM - OPEN FIELD WEEKLY DAILY ENTRY FORM

Bed Number(s)	Crop(s)	Growth Stage	Week of (Mon - Sun)
______________	______________	______________	__________________ to __________________

Worker Name: _______________________________

MONDAY Date: _________________

Item	Morning Entry	Evening Entry
Time of Arrival		
Irrigation Run Done (Yes / No)		
Irrigation Start Time		
Irrigation End Time		
Drip Line Faults (Yes / No - describe)		
Plants with Visible Problems (Yes / No - describe)		
Pest or Disease Signs (Yes / No - describe)		
Mulch Film Damage (Yes / No - describe)		
Inter-row Weed Level (Low / Moderate / High)		
Crop Debris Removed (Yes / No)		
Staking and Ties Checked (Yes / No) - Tomatoes		
Any Other Observation		

Worker Sig (AM): ________________________ Worker Sig (PM): ________________________
Agronomist Signature: ________________________ Notion Updated: Yes / No

TUESDAY Date: _________________

Item	Morning Entry	Evening Entry
Time of Arrival		
Irrigation Run Done (Yes / No)		
Irrigation Start Time		
Irrigation End Time		
Drip Line Faults (Yes / No - describe)		
Plants with Visible Problems (Yes / No - describe)		
Pest or Disease Signs (Yes / No - describe)		

Item	Morning Entry	Evening Entry
Mulch Film Damage (Yes / No - describe)		
Inter-row Weed Level (Low / Moderate / High)		
Crop Debris Removed (Yes / No)		
Staking and Ties Checked (Yes / No) - Tomatoes		
Any Other Observation		

Worker Sig (AM): ________________________ Worker Sig (PM): ________________________
Agronomist Signature: ________________________ Notion Updated: Yes / No

WEDNESDAY Date: ________________

Item	Morning Entry	Evening Entry
Time of Arrival		
Irrigation Run Done (Yes / No)		
Irrigation Start Time		
Irrigation End Time		
Drip Line Faults (Yes / No - describe)		
Plants with Visible Problems (Yes / No - describe)		
Pest or Disease Signs (Yes / No - describe)		
Mulch Film Damage (Yes / No - describe)		
Inter-row Weed Level (Low / Moderate / High)		
Crop Debris Removed (Yes / No)		
Staking and Ties Checked (Yes / No) - Tomatoes		
Any Other Observation		

Worker Sig (AM): ________________________ Worker Sig (PM): ________________________
Agronomist Signature: ________________________ Notion Updated: Yes / No

THURSDAY Date: ________________

Item	Morning Entry	Evening Entry
Time of Arrival		
Irrigation Run Done (Yes / No)		
Irrigation Start Time		
Irrigation End Time		
Drip Line Faults (Yes / No - describe)		
Plants with Visible Problems (Yes / No - describe)		

Item	Morning Entry	Evening Entry
Pest or Disease Signs (Yes / No - describe)		
Mulch Film Damage (Yes / No - describe)		
Inter-row Weed Level (Low / Moderate / High)		
Crop Debris Removed (Yes / No)		
Staking and Ties Checked (Yes / No) - Tomatoes		
Any Other Observation		

Worker Sig (AM): _________________________ Worker Sig (PM): _________________________
Agronomist Signature: _________________________ Notion Updated: Yes / No

FRIDAY Date: _________________

Item	Morning Entry	Evening Entry
Time of Arrival		
Irrigation Run Done (Yes / No)		
Irrigation Start Time		
Irrigation End Time		
Drip Line Faults (Yes / No - describe)		
Plants with Visible Problems (Yes / No - describe)		
Pest or Disease Signs (Yes / No - describe)		
Mulch Film Damage (Yes / No - describe)		
Inter-row Weed Level (Low / Moderate / High)		
Crop Debris Removed (Yes / No)		
Staking and Ties Checked (Yes / No) - Tomatoes		
Any Other Observation		

Worker Sig (AM): _________________________ Worker Sig (PM): _________________________
Agronomist Signature: _________________________ Notion Updated: Yes / No

SATURDAY Date: _________________

Item	Morning Entry	Evening Entry
Time of Arrival		
Irrigation Run Done (Yes / No)		
Irrigation Start Time		
Irrigation End Time		
Drip Line Faults (Yes / No - describe)		

Item	Morning Entry	Evening Entry
Plants with Visible Problems (Yes / No - describe)		
Pest or Disease Signs (Yes / No - describe)		
Mulch Film Damage (Yes / No - describe)		
Inter-row Weed Level (Low / Moderate / High)		
Crop Debris Removed (Yes / No)		
Staking and Ties Checked (Yes / No) - Tomatoes		
Any Other Observation		

Worker Sig (AM): ________________________ Worker Sig (PM): ________________________
Agronomist Signature: ________________________ Notion Updated: Yes / No

SUNDAY Date: ________________

Item	Morning Entry	Evening Entry
Time of Arrival		
Irrigation Run Done (Yes / No)		
Irrigation Start Time		
Irrigation End Time		
Drip Line Faults (Yes / No - describe)		
Plants with Visible Problems (Yes / No - describe)		
Pest or Disease Signs (Yes / No - describe)		
Mulch Film Damage (Yes / No - describe)		
Inter-row Weed Level (Low / Moderate / High)		
Crop Debris Removed (Yes / No)		
Staking and Ties Checked (Yes / No) - Tomatoes		
Any Other Observation		

Worker Sig (AM): ________________________ Worker Sig (PM): ________________________
Agronomist Signature: ________________________ Notion Updated: Yes / No

WEEKLY SUMMARY

Item	Total for the Week
Total Irrigation Runs This Week	
Total Plants Lost - Bed 1	
Total Plants Lost - Bed 2	

Item	Total for the Week
Total Plants Lost - Bed 3	
Total Chemical Applications This Week	
Total Foliar Applications This Week	
Total Harvests This Week	
Any Escalations to Farm Manager (Yes / No)	

Farm Manager Final Weekly Signature: ________________________ Date (Sunday): __________________
Week Status: Satisfactory / Unsatisfactory (circle one)
Farm Manager Comments: ____

FORM B - FERTIGATION AND NUTRIENT LOG

WGNI FARM - FERTIGATION AND NUTRIENT LOG

Bed Number	Crop	Week of
______________	______________	__________________

Date	Fertiliser / Nutrient Product Used	Rate Applied	Method (Drip / Foliar)	Agronomist Signature
Monday				
Tuesday				
Wednesday				
Thursday				
Friday				
Saturday				
Sunday				

Farm Manager Review Signature: __________________________ Date: ___________________

FORM C - WEEKLY CLEANING AND INSPECTION SIGN-OFF

WGNI FARM - OPEN FIELD WEEKLY INSPECTION SIGN-OFF

Date	Inspector
_____________ (Every Saturday)	Farm Manager alongside Agronomist

Area	Standard	Result	Action Required
Bed 1 - Plant health	Uniform, no widespread disease or wilting	Pass / Fail	
Bed 2 - Plant health	Uniform, no widespread disease or wilting	Pass / Fail	
Bed 3 - Plant health	Uniform, no widespread disease or wilting	Pass / Fail	
Drip lines and emitters - all beds	All functioning, no leaks or dry sections	Pass / Fail	
Mulch film - all beds	No tears, lifting, or displacement	Pass / Fail	
Inter-row walkways - all beds	Weeds below 10 cm, no standing water	Pass / Fail	
Staking and ties - tomatoes	All plants supported, no tight ties	Pass / Fail	
Field debris removal	All crop debris removed from beds and walkways	Pass / Fail	
Drip line filters	Flushed and clean	Pass / Fail	
Tools and equipment	Cleaned, disinfected, and stored	Pass / Fail	
Chemical store	Locked, labelled, organised	Pass / Fail	
Field perimeter	No encroachment or security concerns	Pass / Fail	
Orchard demarcation	Boundary markers intact	Pass / Fail	
Orchard base circles	Weed-free and mulched	Pass / Fail	
Orchard inter-row grass	Below 15 cm	Pass / Fail	
Orchard tree health	No wilting, pest activity, or stake damage	Pass / Fail	

Overall Result	Areas to Redo
Pass / Fail (circle one)	

Re-inspection Completed	Date
Yes / No	_________________

Agronomist Signature: ________________________ Date: _________________
Farm Manager Signature: ________________________ Date: _________________ Time: _________________

FORM D - CHEMICAL AND FOLIAR APPLICATION RECORD

WGNI FARM - CHEMICAL AND FOLIAR APPLICATION RECORD

Location (Bed / Orchard)	Crop / Tree	Date	Applied By
______________	______________	______________	______________

Item	Details
Type of Application	Pesticide / Fungicide / Herbicide / Foliar Feed (circle one)
Pest, Disease, Deficiency, or Weed Being Treated	
Product Name	
Active Ingredient or Nutrient Formulation	
Previous Product Used for Same Issue (rotation check)	
Dilution Rate	
Volume Prepared	
Method of Application (Foliar / Soil Drench / Herbicide inter-row)	
Rows or Areas Treated	
Time of Application	
Re-entry Interval (from product label)	
Re-entry Permitted From (date and time)	
Protective Gear Worn (Yes / No)	
Product Returned to Store After Use (Yes / No)	

Worker Signature: ________________________ Date: _________________
Agronomist Authorisation Signature: ________________________ Date: _________________
Farm Manager Awareness Signature: ________________________ Date: _________________

FORM E - HARVEST RECORD FORM

WGNI FARM - OPEN FIELD HARVEST RECORD

Bed Number	Crop	Harvest Date	Harvest Number This Cycle
______________	______________	______________	______________

Grade	Quantity (kg)	Units / Count	Destination
Grade A (Full size, no blemish)			
Grade B (Minor imperfection, marketable)			
Reject (Damaged, diseased, substandard)			
TOTAL			

Reject Disposal Method: ____
Observations During Harvest: ____
Harvested By: ________________________ Confirmed By (Agronomist): ________________________

Worker Signature: ________________________ Date: __________________
Agronomist Signature: ________________________ Date: __________________

FORM F - ORCHARD MAINTENANCE AND OBSERVATION LOG

WGNI FARM - ORCHARD MAINTENANCE AND OBSERVATION LOG

Week of	Agronomist	Worker
__________________ to __________________	____________________________	____________________________

Date	Activity Carried Out	Trees Affected / Area	Observation or Notes	Agronomist Signature
Monday				
Tuesday				
Wednesday				
Thursday				
Friday				
Saturday				
Sunday				

WEEKLY ORCHARD SUMMARY

Item	Details
Total Trees Inspected This Week	
Any Trees with Pest or Disease Signs (Yes / No - describe)	
Chemical Applications This Week (Yes / No - describe on Form D)	
Pruning Carried Out (Yes / No - describe)	
Mulch Replenished (Yes / No)	
Inter-row Grass Cut (Yes / No)	
Base Circle Weeded (Yes / No)	
Any Trees Requiring Urgent Attention	

Farm Manager Review Signature: ________________________ Date: __________________

DOCUMENT AUTHORISATION

FIELD	DETAIL
Document Title	Open Field and Orchard Operations Standard Operating Procedure
Organisation	WGNI Farm, Abeokuta, Nigeria
Version	1.0
Date Issued	April 2026
Prepared By	Farm Manager / Agronomist
Authorised By	CEO, WGNI Farm
Next Review Date	October 2026

This SOP is authorised by the CEO of WGNI Farm. It covers all currently active open field seed beds and the orchard. Every new bed or orchard section added in the future automatically falls under the full scope of this document. This SOP is subject to periodic review.